Prym census observation integrity

TASK-20261025-5145ac | EXP-ECDLP-d6d8e8 | RUN-ECDLP-ea49a6

Assigned-joint integrity findings only. No scientific hypothesis verdict or state transition.

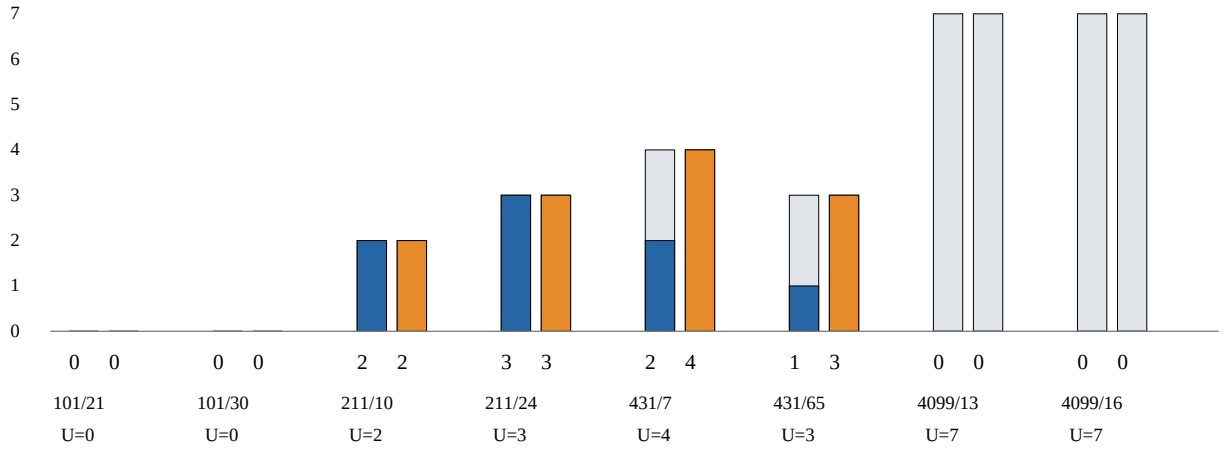
Blind derivation sealed 2026-10-25T03:44:38.004075+00:00. SHA256 0751c0c2771e291a363d92c99d5b16cdce27a2a340abe6080e150394b418db84.

Fresh code checked 1,867,776 attempted outcomes and digests, 4,257 E1 source equations, all 16 projected and augmented ranks, pivots, solutions, kernels and rank checkpoints. Fixed principal/E2 sample: 144 accepted rows.

p	B	r	U	sample rows	full rows	sample rank	full rank
101	21	107	0	31	1244	0	0
101	30	97	0	40	1506	0	0
211	10	199	2	17	811	2	2
211	24	197	3	6	319	3	3
431	7	433	4	2	109	2	4
431	65	443	3	1	171	1	3
4099	13	4021	7	0	0	0	0
4099	16	4111	7	0	0	0	0

Exact projected rank after public anchors

Blue: sample64; orange: full3584; grey: U. Exact dimensions, n=8 fixtures.



Labels p/B; sources a=1..64, seed0. RUN-ECDLP-ea49a6.

Exact finite counts over 8 fixtures and 16 cells; no population interval. U=0 cells are all-public calibration. Seed0 is ordering nuisance.

Findings by assigned joint

fixture_public_source_projection: holds

Eight fixture order/prime/generator/base recounts and all 64 declared public images each, signed canonical anchor maps/U; UID/EC1 numerical identity before seal, exact serialization hashes after seal.

U=[0,0,2,3,4,3,7,7]; identical projection in both arms; no truth-log table computed.

Limits: Exact identity serialization deliberately deferred to phase2.

complete_support_and_alias_law: holds

All 1024 ordered source blocks and 1,867,776 descriptors; fresh lift/rejection/alias/interpolation/residual/acceptance outcome and attempt-digest regeneration; all A/A witness hashes and transported E1 equations; sampled canonical equations nested in full up to global sign.

All stored outcomes, source checkpoints, candidate counts and accepted row locations match. No rejected work omitted; all fixed work continues beyond first full rank.

Limits: Canonicalizer implements the declared finite geometric deck/hyperelliptic orbit, not a global equivalence theorem.

source_witness_and_exact_rank: holds

All 4257 public E1 equations and 16 matrices/rank/augmented rank/pivots/solutions/right-kernel/progress/first-full certificates; 144 fixed principal/E2 witness samples (first 4,last 4,SHA-ranked 8,deduplicated; all below 16).

Exact matrix values match stored results; full ranks by fixture 0,0,2,3,4,3,0,0 versus U 0,0,2,3,4,3,7,7.

Limits: Explicit E2 norm and full principal witness check is the preregistered deterministic sample, not every accepted row.

native_custody_and_cost_scope: holds

Approved spec/plan/source/input committed bindings, native claim/source/clean-tree receipt state,19 logical outputs plus archive record, native two-pointer/stored/raw hashes and canonical manifest/custody, own cache readback, partial-counter scope.

Committed snapshot bytes and every digest bridge match; all 20 logical snapshot paths accounted. Source commit reachable in cached origin/lane/prym-census-run and origin/lane/prym-rank-census refs. Native published:false is recorded runner publication state; archival commits supply durable bytes.

Limits: Own remote GET and independent operation-counter recomputation not performed. No complete weighted S, process-group memory trace, prescribed-computation proof or settlement.

Controls, custody and limits

All 11 reviewer negative mutations are rejected: scalar a+1, altered U, altered pivot, missing attempted status, changed raw digest, changed stored digest, omitted infinity term, repeated residual root, wrong atom sign, post-hoc locator and unknown tuple source. Exact mutations and rejection messages are retained in verification.json.

Every 4257 accepted interpolation/residual/row is recomputed in the complete attempted-stream replay. Explicit complete principal/E2 checks use first 4, last 4 and 8 SHA-ranked accepted rows per cell, deduplicated; all rows if fewer 16. Domain: TASK-20261025-5145ac-principal-v1. Existing checker receipt is corroboration only.

Finite panel of eight deterministically selected toy curves over four prime fields; seed0 is ordering nuisance, not an independent curve.

Explicit full principal-divisor/E2 audit covers 144 deterministic accepted rows; every 4257 accepted interpolation/residual/row and E1 equation is replayed. Existing native checker reports every principal witness verified; that receipt is borrowed corroboration.

Own custody readback is from the public cache only. The immutable producer/custody records assert fresh remote GET; no independent reviewer remote GET or remote availability conclusion is supplied.

Operation counters were not independently recomputed; input search/checker/storage costs remain separately scoped, with no complete weighted S or execution proof.

UID/EC1 exact hash serialization checked after blind seal; mathematical input/order/base/public projection derived before unblinding.

Generic independence checker cannot represent this exact same-task phased blind plan; report retains complete source disclosure and diagnostic rather than claiming its static whole-session gate passed.

Producer diagnostics: 97.490447 wall seconds, 97.471312 CPU seconds, 49266688 peak self RSS bytes. Existing checker: 73.964993 wall seconds, 73.951064 CPU seconds, 58650624 peak self RSS bytes.

Review attestation: separate reviewer session; no producer authorship; no sibling report read; phase1 blind-from respected through sealed derivation; phase2 deliberately unblinded. Complete source-read list, per-joint scope and deviations: report.json and report.md. Exact output/recomputation records: verification.json; executable reviewer source: review.py.

No external canonical graph changes: task authorizes finite observation-integrity review only. Internal source provenance: reviewer TASK-20261025-5145ac read committed snapshot a6fe1b471 and bound sources. No new scientific trial, credential access, source edit or state transition.